ROLEEDGE · CONSULTING PREP

Case Math Workbook

Build Lightning-Fast Accuracy for Consulting & Finance Interviews

15	40	Print
SECTIONS	PRACTICE QUESTIONS	WITH SPACE TO WRITE

How to use this workbook: print it, work through it in order, and write in the spaces. Every exercise has room for your answers, and you can print extra copies of any page.

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From **RoleEdge**, founded by Mayank Kapoor, former Manager at Bain & Company and Warwick MBA. Free to use and share.

Why Case Math Is Critical in Interviews

Consulting and finance interviews demand **structured thinking and numerical accuracy.** Even if your structure is brilliant, sloppy math can ruin your impression. Interviewers want to see:

- Comfort with numbers under time pressure.
- Ability to approximate without paralysis.
- Logical step-by-step explanation.

EXERCISE

Think of the last time you froze on a math question. What went wrong—panic, accuracy, or communication?

Core Math Skills Every Candidate Needs

Before solving complex cases, master the basics:

- Multiplication & division shortcuts.
- Ratios and proportions.
- Percentage increase/decrease.
- Compounding growth.
- Weighted averages.

QUICK CHECK

What's 25% of 480? Did you solve it in your head or on paper?

Percentages, Ratios, and Growth Rates

EXAMPLE

A company grows revenue from ₹100 Cr to ₹120 Cr: growth rate = 20%.

If profits are ₹20 Cr on ₹100 Cr revenue, margin = 20%.

Ratio of fixed to variable cost = 2:3.

EXERCISE

A firm's revenue grows from $200M to $260M in 2 years. What's the CAGR?

Profitability Calculations & Breakevens

Profit = Revenue – Costs.

Breakeven = Fixed Costs ÷ Contribution Margin.

EXAMPLE

Fixed Costs = ₹50 Cr.

Selling Price = ₹500.

Variable Cost = ₹300.

Contribution Margin = ₹200.

BEP Units = 50 Cr ÷ 200 = 25 lakh units.

EXERCISE

Calculate breakeven for a SaaS firm with $2M fixed costs and $40 contribution margin.

Market Sizing Math in Action

EXAMPLE

Estimate taxi revenue in Mumbai.

- 2M daily rides.
- Avg fare = ₹300.
- Revenue per day = 2M × ₹300 = ₹60 Cr.
- Annual revenue = ₹60 Cr × 365 ≈ ₹22,000 Cr.

EXERCISE

Estimate annual toothpaste revenue in India if 4B tubes are sold at ₹50 each.

Pricing & Revenue Modeling

Pricing strategy often needs math:

- If price ↑ 10% and demand ↓ 5%, net revenue ↑ ~4.5%.
- Bundling: 3 items for ₹99 instead of ₹40 each (₹120) cuts the price by 17.5%, which changes the margin.

EXERCISE

A firm sells 10,000 units at ₹500 each, and each unit costs ₹300. What are revenue and profit?

Cost Structures & Margin Analysis

Example:

Revenue = ₹100 Cr.

COGS = ₹60 Cr.

Gross Margin = 40%.

OpEx = ₹20 Cr.

EBIT Margin = 20%.

EXERCISE

If revenue is $80M and costs are $68M, what's the profit margin?

ROI & Investment Calculations

ROI = (Net Gain ÷ Investment).

NPV/IRR used in finance cases.

EXAMPLE

Invest ₹100 Cr.

Gain ₹140 Cr in 2 years.

ROI = 40%. CAGR ≈ 18%.

EXERCISE

A startup investor puts in $2M and exits at $3.5M after 3 years. What's the CAGR?

Capacity & Utilization Math

EXAMPLE

A factory capacity = 100,000 units/month. Running at 75% → produces 75,000.

EXERCISE

A call centre has 500 agents. Each can handle 20 calls a day, and utilisation is 80%. How many calls a day?

Common Shortcuts for Quick Mental Math

- % of multiples of 10 (12% of 250 = 25 + 5 = 30).
- CAGR = (End ÷ Start)^(1/n) – 1. For two years, take the square root.
- Rule of 72 for doubling time: 72 ÷ growth rate.

EXERCISE

Estimate the doubling time at 12% CAGR.

Practice set 1: Basic math

1. Costs rise 8% on a base of ₹500 Cr. What is the new cost?

2. Revenue grows 25% from $80M. What is the new revenue?

3. 180 salespeople are split between two regions in a 4:5 ratio. How many are in each?

4. What is 35% of 640?

5. Revenue falls from ₹250 Cr to ₹200 Cr. What is the percentage change?

6. A price rises from ₹80 to ₹100. What is the percentage increase?

7. What is 18% of 450?

8. A brand sells 2.4M units in a 16M-unit market. What is its market share?

9. 60% of sales earn a 20% margin and 40% earn 35%. What is the blended margin?

10. What is 7.5 × 1,200?

Practice set 2: Profit and pricing

1. Revenue is ₹120 Cr and total costs are ₹96 Cr. What is the profit margin?

2. Fixed costs are ₹30 Cr, the price is ₹1,000 and the variable cost is ₹600. How many units to break even?

3. Price rises 10% and volume falls 10%. What happens to revenue?

4. A café sells 400 coffees a day at ₹180. Each cup costs ₹70 to make. What is the daily contribution?

5. Revenue is $50M and the gross margin is 40%. What is the cost of goods sold?

6. A SaaS firm has 2,000 customers paying $500 a month. What is its annual revenue?

7. The price is cut 20%. By how much must volume grow to keep revenue flat?

8. Contribution is $25 a unit, fixed costs are $1M and the profit target is $250K. How many units?

9. A SaaS startup adds 100 clients at $1,000 a year and spends $50,000 to serve them. What is the profit margin?

10. Revenue is ₹200 Cr, COGS ₹110 Cr and operating costs ₹50 Cr. What is the EBIT margin?

Practice set 3: Market sizing and capacity

1. A city has 10M people. 30% drink coffee daily, 1.5 cups each. How many cups a day?

2. A plant can make 1.2M units and runs at 70% capacity. How many units does it make?

3. A warehouse ships 12,000 orders a day with 60 packers. How many orders per packer?

4. A 200-seat plane flies 40 times a day at 80% load. How many passengers a year?

5. A hotel has 250 rooms at 72% occupancy and ₹6,000 a night. What is its annual room revenue?

6. Of 1.4B people, 25% buy a product twice a year at ₹250. What is the market size?

7. A machine makes 90 units an hour, 2 shifts of 8 hours, 300 days a year. What is its annual output?

8. A gym has 1,500 members paying ₹2,000 a month. What is its annual revenue?

9. 2M households, 90% own a TV and replace it every 8 years. How many TVs are sold a year?

10. A plant at 60% utilisation makes 30,000 units. What is its capacity?

Practice set 4: Investment and growth

1. A PE firm invests $50M and exits at $120M after 5 years. What is the CAGR?

2. A project costs ₹50 Cr and earns ₹15 Cr a year. What is the payback period?

3. $10M invested returns $4M a year. What is the simple payback?

4. Using the rule of 72, how long does revenue growing 9% a year take to double?

5. At a 10% discount rate, what is ₹121 received in 2 years worth today?

6. Revenue of $100M grows 20% a year. What is it after 3 years?

7. A market grows from ₹400 Cr to ₹900 Cr in 4 years. What is the CAGR?

8. Capacity rises from 1M to 1.3M units while demand is 1.2M. What is utilisation now?

9. $5M grows at 8% a year. Roughly what is it worth after 9 years?

10. Revenue doubles from $100M to $200M in 3 years. What is the CAGR?

Answers

Check your working once you've tried each question. Rounded answers are fine in an interview if you say you're rounding.

Question	Answer
Core math skills	25% of 480 = 120
Growth rates	CAGR ≈ 14% (1.3 over 2 years: √1.3 ≈ 1.14)
Breakevens	$2M ÷ $40 = 50,000 customers
Market sizing	4B × ₹50 = ₹20,000 Cr
Pricing	Revenue ₹50 lakh, profit ₹20 lakh
Margins	$12M ÷ $80M = 15%
ROI	CAGR ≈ 21% (1.75 over 3 years)
Capacity	500 × 20 × 80% = 8,000 calls
Shortcuts	72 ÷ 12 ≈ 6 years
Set 1, Q1	₹540 Cr
Set 1, Q2	$100M
Set 1, Q3	80 and 100
Set 1, Q4	224
Set 1, Q5	−20%
Set 1, Q6	25%
Set 1, Q7	81
Set 1, Q8	15%
Set 1, Q9	26%
Set 1, Q10	9,000
Set 2, Q1	20%
Set 2, Q2	7.5 lakh units
Set 2, Q3	Falls 1%
Set 2, Q4	₹44,000
Set 2, Q5	$30M
Set 2, Q6	$12M
Set 2, Q7	25%
Set 2, Q8	50,000
Set 2, Q9	50%

Question	Answer
Set 2, Q10	20%
Set 3, Q1	4.5M
Set 3, Q2	840,000
Set 3, Q3	200
Set 3, Q4	About 2.3M (2,336,000)
Set 3, Q5	About ₹39.4 Cr
Set 3, Q6	₹17,500 Cr
Set 3, Q7	432,000 units
Set 3, Q8	₹3.6 Cr
Set 3, Q9	About 225,000
Set 3, Q10	50,000 units
Set 4, Q1	About 19%
Set 4, Q2	About 3.3 years
Set 4, Q3	2.5 years
Set 4, Q4	About 8 years
Set 4, Q5	₹100
Set 4, Q6	$172.8M
Set 4, Q7	About 22%
Set 4, Q8	About 92%
Set 4, Q9	About $10M (rule of 72)
Set 4, Q10	About 26%

My notes

NEXT STEP

Want someone to check your work?

Bring this workbook to a free call with RoleEdge. We'll go through your answers and help you plan what comes next.

Book a free call at roleedge.com/book